Seat No._____

# GANPAT UNIVERSITY

## B.Sc. (C.A. & I.T.) SEM-III EXAMINATION (CBCS)

## U 1/3A1DS: DATA STRUCTURE

**Time: 3 Hours]** Nov - Dec 2022 **[Total Marks: 60**

**Instructions:**

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

## SECTION-I

**1. Answer the following: (Any 5)** **[30]**

1. Distinguish Primitive & Non-primitive Data Structure with an example.
2. What is data structure? Difference between static and dynamic data structure with Example.
3. What is Stack? List out the operations of Stack. Write algorithm to Push an It into the Stack.
4. What is simple Queue? Write an algorithm to Insert and Delete Operations of simple Queue.
5. Explain AVL tree with example.
6. What is DFS? Explain the algorithm for DFS.
7. Write down the algorithm of Quick sort, and short the following sequence Number using quick sort: 5, 3,8, 1, 4, 6 , 2 and 7

## SECTION-II

**2. Answer the following: (Any 2)** **[12]**

1. List out the stack applications, and Write a recursive algorithm that finds the Fibonacci no.
2. Describe singly linked list, write an algorithm for search operation.
3. What is Tree? Explain types of Tree data structure with example.

**3. Answer the following: (Any 2)** **[12]**

1. Write down the Pre-Order algorithm and example for BST.
2. Explain BFS traversal and write algorithm for BFS.
3. Explain types of linked list data structure and explain any one linked list in detail.

**4. Answer the following: (Any 1)** **[06]**

1. What is sorting? Explain the Bubble Sort with suitable Example.
2. What is queue? Explain types of queue with example.

**END OF PAPER**

Seat No._____

# GANPAT UNIVERSITY
## B.Sc. (CA & IT) SEM-III EXAMINATION

## U13A2OCP: OBJECT ORIENTED CONCEPTS AND PROGRAMMING

Nov - Dec 2022

**Time: 3 Hours]** **[Total Marks: 60**

**Instructions:**

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

### SECTION-I

**1. Answer the following: (Any Five)** **(30)**

1. Explain Features of OOP.
2. How StringBuffer class is different from String class? Discuss any two methods that is not available in String class.
3. What is Throwable? Explain try, catch finally, throw and throws.
4. What is method overloading? Explain with example
5. Explain with example Vector class.
6. What is the use of Lambda expression? In Which Version of jdk Oracle introduced this features? Explain in brief block lambda expression.
7. Explain phaser in detail.

### SECTION-II

**2. Answer the following: (Any Two)** **(12)**

1. Why java is platform independent language? Explain JDK,JRE, JRE, JIT
2. Which are the ways to implements thread in java? Explain with example.
3. Explain access specifier in java.

**3. Answer the following: (Any Two)** **(12)**

1. How to create object of Calendar class? List out any six constants of Calendar.
2. Explain method reference in lambda expression.
3. Differentiate HashTable and Vector. Explain HashTable in detail

**4. Answer the following: (Any One)** **(06)**

1. Discuss fork/join framework
2. Explain executer in detail.

**END OF PAPER**

Seat No._____

# GANPAT UNIVERSITY
## B.Sc. (CA & IT) SEM-III EXAMINATION

## U13A3DMS: DATABASE MANAGEMENT SYSTEM

Nov - Dec 2022

**Time: 3 Hours]** **[Total Marks: 60**

**Instructions:**

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

### SECTION-I

**1. Answer the following: (Any Five)** **(30)**

1. Explain Advantages of Database Management System.
2. Discuss Dr. E. F. Codd's rules for RDBMS.
3. What is E-R Model? Explain components of E-R Model.
4. Explain 2NF and 3NF.
5. Explain Oracle Data Types.
6. What is Join? Explain with example and list out types of it.
7. What is Sequence? Explain keywords and parameters of Sequence.

### SECTION-II

**2. Answer the following: (Any Two)** **(12)**

1. Elaborate Components of Database System.
2. Discuss Role of Database Administrator.
3. Explain Fifth Normal Form in detail.

**3. Answer the following: (Any Two)** **(12)**

1. What is Constraints? List out types of it and discuss any two Constraints.
2. Elaborate Group by and Having clause in SQL.
3. Summarize Subquery in SQL.

**4. Answer the following: (Any One)** **(06)**

1. List out types of index and explain it.
2. Short note on Granting and Revoking permissions.

**END OF PAPER**

Enrollment No._____

# GANPAT UNIVERSITY
## B.Sc. (CA & IT) SEM-III EXAMINATION

### U13A4SAD: SYSTEM ANALYSIS AND DESIGN

Nov - Dec 2022

[Time: 3 Hours] [Total Marks: 60]

**Instructions:**

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

### SECTION-I

**1. Answer the following: (Any Five out Seven)** **(30)**

1. Who is System Analyst? Discuss the role of system analyst.
2. Write a short note on structured English.
3. Discuss the fact finding techniques: - Interviews and Questionnaires.
4. Discuss the difference between physical DFD vs. logical DFD.
5. Draw as well as explain elements of DFD in detail.
6. Explain principles of code design in brief.
7. Draw DFD for Online Examination system for case study.

### SECTION-II **(30)**

**2. Answer the following: ( Any Two)** **(12)**

1. Discuss the element of the system.
2. Explain characteristics of system in brief.
3. Discuss seven phases of the System Development Life Cycle.

**3. Answer the following: (Any Two)** **(12)**

1. Discuss guidelines for Form design.
2. Discuss difference between White Box Testing Vs. Black Box Testing.
3. Discuss Level of Testing in detail.

**4. Answer the following: (Any One)** **(6)**

1. Discuss Case Study of "College Management System".
2. Discuss Case Study of "Hotel Management System".

**END OF PAPER**

Seat No._____

# GANPAT UNIVERSITY
## B.Sc. (CA & IT) SEM-III EXAMINATION
Nov - Dec 2022
## U13A5OR: OPERATION RESEARCH

**Time: 3 Hours]** **[Total Marks: 60**

**Instructions:**

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

### SECTION-I

**1. Answer the following: (Any Five)** **(30)**

1. Discuss the structure of Linear Programming Problem.
2. Solve the following LPP by using **Graphical method.**

   Max $Z = 2x_1 + 3x_2$

   S.T.C $x_1 + x_2 \leq 30$

   $x_2 \geq 3$

   $0 \leq X_2 \leq 12$

   $X_1 - X_2 \geq 0$

   and $x_1, x_2 \geq 0$
3. Discuss the step of VAM method.
4. Consider the problem of assigning four sales person to four different sales region as shown in table such that the total sales is maximized

| | R1 | R2 | R3 | R4 |
|---|---|---|---|---|
| S1 | 20 | 44 | 24 | 28 |
| S2 | 32 | 36 | 44 | 20 |
| S3 | 48 | 40 | 24 | 36 |
| S4 | 32 | 28 | 48 | 40 |

   The cell entries represent annual sales figures in lakhs of rupees. Find the optimal allocation of the sales persons to different region.
5. Define Terms: a) Unbalanced Transportation Problem
   b) Who invented the term of Operations Research?
   c) What is RIM Condition?
   d) MODI Method is also known as _____.
   e) Which method used to find out Penalty?
   f) M + N - 1 = Allocation is known as_____.
6. Discuss mathematical model of Assignment Problem.
7. Discuss the step of Processing N Jobs on 2 Machines.

### SECTION-II

**2. Answer the following: (Any Two)** **(12)**

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# GANPAT UNIVERSITY
## B.Sc. (CA & IT) SEM-III EXAMINATION
Nov - Dec 2022
## U13B6EDM: ENVIRONMENT DISASTER MANAGEMENT

**Time: 3 Hours]** **[Total Marks: 60**

**Instructions:**

1. Figures to the right indicate full marks
2. Each section should be written in a separate answer book
3. Be precise and to the point in your answer

### SECTION-I

**1. Answer the following: (Any Five)** **(30)**

a) What is Environment? Define the Scope of Environment.

b) **Define Terms:** Ecology and Ecosystem.

c) What is Water Pollution? Define different effects of Water Pollution.

d) What is Air Pollution? Describe classification of Air Pollution.

e) Write a short note on Global Warming & Green Houses Effects

f) What is Disaster? Discuss Disaster Management Cycle.

g) Explain the role of Insurance Company in Rehabilitation.

### SECTION-II

**2. Answer the following: (Any Two)** **(12)**

a) Write a short note on importance of Environment Studies.

b) Discuss the classification and function of Ecosystem.

c) **Define Terms:** Producers, Consumers and Decomposer.

**3. Answer the following: (Any Two)** **(12)**

a) What is Noise Pollution? Define sources of Noise Pollution.

b) What is Acid Rain? Discuss effect and solution of Acid Rain

c) What is Hazard? Explain types of Hazards.

**4. Answer the following: (Any One)** **(06)**

a) Describe the role of Police in Rehabilitation.

b) Define the role of Non Government in Rehabilitation.

**END OF PAPER**